



The Funk–Radon transform

A classical problem from tomography and its computational inversion

TU Chemnitz, Faculty of Mathematics, PhD seminar

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Michael Quellmalz

TU Chemnitz, Faculty of Mathematics, PhD seminar

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1. Introduction

Definition

History

2. The Funk–Radon transform

Spherical harmonics as eigenfunctions of the Funk–Radon transform

Sobolev spaces

3. Inversion of the Funk–Radon transform

Discrete data

Noisy data

Bias–variance tradeoff

Optimal mollifiers

4. Generalizations of the Funk–Radon transform

Convolution operators

Means along circles on the sphere

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Definition

- ▶ Sphere $\mathbb{S}^2 = \{\xi \in \mathbb{R}^3 : \|\xi\| = 1\}$
- ▶ Function $f : \mathbb{S}^2 \rightarrow \mathbb{C}$

Funk–Radon transform

The Funk–Radon transform (a.k.a. spherical Radon transform, Funk transform, Minkowski–Funk transform) of f is given by

$$\mathcal{R}f(\xi) := \frac{1}{2\pi} \int_{\xi^\perp} f(\eta) \, ds(\eta), \quad \xi \in \mathbb{S}^2.$$

- ▶ $\xi^\perp := \{\eta \in \mathbb{S}^2 : \langle \eta, \xi \rangle = 0\}$... great circle perpendicular to ξ
- ▶ ds ... arc length on $\xi^\perp$

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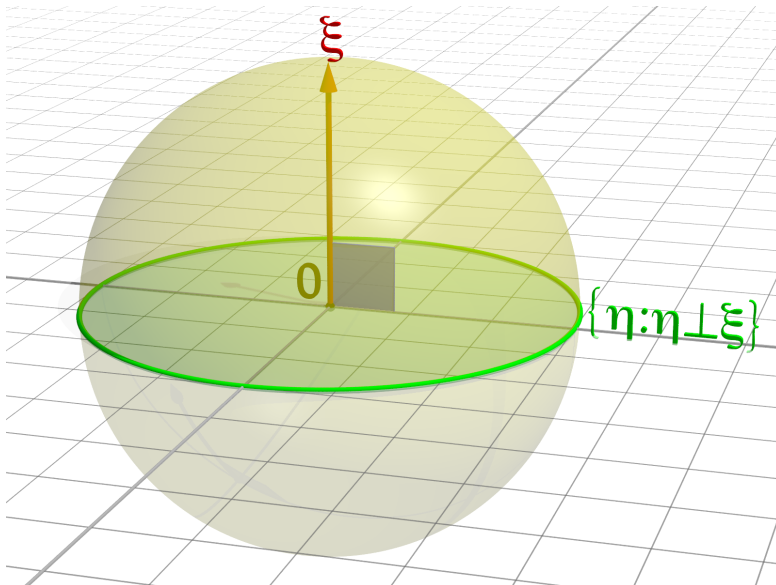
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A few remarks on this definition

- ▶ Recall the Funk–Radon transform

$$\mathcal{R}f(\xi) := \frac{1}{2\pi} \int_{\xi^\perp} f(\eta) \, ds(\eta), \quad \xi \in \mathbb{S}^2$$

- ▶ f continuous $\Rightarrow \mathcal{R}f$ continuous
- ▶ Normalization: $f \equiv 1 \Rightarrow \mathcal{R}f \equiv 1$
- ▶ $\mathcal{R}f$ is even, i.e. $\mathcal{R}f(\xi) = \mathcal{R}f(-\xi)$
- ▶ If f is odd, i.e. $f(\xi) = -f(-\xi)$, then $\mathcal{R}f$ vanishes
- ▶ $\mathcal{R}$ commutes with rotations

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A geometric interpretation

(Lutwak, 1988)

- ▶ A **star-shaped body** $K \subset \mathbb{R}^3$ with center 0 is a compact set with the property: If $\xi \in K$ then the line segment $[0, \xi] \in K$
- ▶ A star-shaped body is characterized by its **radial function**

$$\rho_K(\xi) = \max\{\lambda \geq 0 : \lambda\xi \in K\}, \quad \xi \in \mathbb{S}^2$$

- ▶ The **intersection body** IK of K is the star-shaped body with the radial function

$$\rho_{IK}(\xi) = \text{vol}_2(K \cap \xi^\perp), \quad \xi \in \mathbb{S}^2$$

Theorem

$$\rho_{IK} = \pi \mathcal{R} \rho_K$$

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Why so many names for the Funk–Radon transform?

Hermann Minkowski, 1904: *About bodies of constant width*

- Integral over a great circle was used in a proof

Paul Funk, 1911: *Über Flächen mit lauter geschlossenen geodätischen Linien*

Wanted to reconstruct a function $f : \mathbb{S}^2 \rightarrow \mathbb{R}$ on the sphere

Given $\mathcal{R}f$ which he called **circle-integral function** of f
("Kreis-Integralfunktion")

Johann Radon, 1918: *Über die Bestimmung von Funktionen durch ihre Integralwerte längs gewisser Mannigfaltigkeiten*

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Both the great circles on the sphere and the lines in the plane are **geodesics**

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- Every function $f \in L^2(\mathbb{S}^2)$ can be written as a Fourier series

$$f = \sum_{n=0}^{\infty} \sum_{k=-n}^n \hat{f}(n, k) Y_n^k$$

with the Fourier coefficients $\hat{f}(n, k) := \int_{\mathbb{S}^2} f \overline{Y_n^k}$

- Spherical harmonics Y_n^k are harmonic polynomials of degree n

Eigenvalue decomposition of the Funk–Radon transform (Funk, 1911)

$$\mathcal{R}f = \sum_{n=0}^{\infty} \sum_{k=-n}^n \hat{\mathcal{R}}(n) \hat{f}(n, k) Y_n^k$$

where

$$\hat{\mathcal{R}}(n) = \begin{cases} (-1)^{n/2} \frac{(n-1)!!}{n!!}, & n \text{ even} \\ 0, & n \text{ odd.} \end{cases}$$

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Sobolev spaces

- ... are used to describe the smoothness of functions

Definition

For $s \geq 0$ the Sobolev space $H^s(\mathbb{S}^2)$ is defined as the completion of the space of all spherical polynomials with respect to the norm

$$\|f\|_s^2 := \sum_{n=0}^{\infty} \sum_{k=-n}^n \left| \hat{f}(n, k) \right|^2 \left(n + \frac{1}{2} \right)^{2s}.$$

- $H^0(\mathbb{S}^2) = L^2(\mathbb{S}^2)$
- $H^s(\mathbb{S}^2) \hookrightarrow H^t(\mathbb{S}^2)$ compact for $s > t$
- $H^s(\mathbb{S}^2) \hookrightarrow C(\mathbb{S}^2)$ for $s > 1$ (Sobolev embedding theorem)

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The Funk–Radon transform in Sobolev spaces

- ▶ $H_e^s(\mathbb{S}^2)$... Sobolev space of even functions

Theorem

Let $s \geq 0$. The Funk–Radon transform

$$\mathcal{R} : H_e^s(\mathbb{S}^2) \rightarrow H_e^{s+1/2}(\mathbb{S}^2)$$

is continuous, bijective and open (i.e. its inverse is continuous).

Proof.

Via the eigenvalues and Stirling's formula



- ▶ The Funk–Radon transform is a smoothing operator
- ▶ $\mathcal{R} : H_e^s(\mathbb{S}^2) \rightarrow H_e^s(\mathbb{S}^2)$ is a compact operator

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The inversion problem

We have $g = \mathcal{R}f \in H_e^{s+1/2}(\mathbb{S}^2)$

We want $f \in H_e^s(\mathbb{S}^2)$

The eigenvalue decomposition yields

$$f = \mathcal{R}^\dagger g = \sum_{\substack{n=0 \\ 2 \mid n}}^{\infty} \sum_{k=-n}^n \frac{\hat{g}(n, k)}{\hat{\mathcal{R}}(n)} Y_n^k$$

Our next steps

1. Sample values of the function g on a discrete set
2. Noisy data $g = \mathcal{R}f + \varepsilon$

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How to handle discrete data

- Idea: Use **quadrature rule** to compute

$$\hat{g}(n, k) = \int_{\mathbb{S}^2} g \overline{Y_n^k} \approx \frac{1}{M} \sum_{m=1}^M g(\xi_m) \overline{Y_n^k(\xi_m)}$$

- Usual orthogonal projection of degree N

$$\mathcal{P}_N g = \sum_{n=0}^N \sum_{k=-n}^n \hat{g}(n, k) Y_n^k$$

- Define **hyperinterpolation** of degree N

$$\mathcal{L}_N g = \sum_{n=0}^N \sum_{k=-n}^n \left(\frac{1}{M} \sum_{m=1}^M g(\xi_m) \overline{Y_n^k(\xi_m)} \right) Y_n^k$$

- So we compute

$$\mathcal{R}^\dagger \mathcal{L}_N g.$$

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Hyperinterpolation

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- ▶ $\mathcal{L}_N g$ is a spherical polynomial of degree $\leq N$
- ▶ Call $\mathcal{L}_N$ **exact** if it's a projection, i.e. $\mathcal{L}_N^2 = \mathcal{L}_N$
- ▶ If the quadrature is exact of degree $2N$, then $\mathcal{L}_N$ is exact
- ▶ For simplicity, we use **equi-weight quadrature**
- ▶ If such quadrature is exact, the set of nodes is called **spherical design**
- ▶ Spherical designs exist with $M \sim N^2$
- ▶ Even if it's exact, $\mathcal{L}_N$ is **not** an interpolation (for $N \geq 3$)

Hyperinterpolation

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We have $g = \mathcal{R}f \in H_e^{s+1/2}(\mathbb{S}^2)$

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$$\psi(t) := \sum_{n=0}^{\infty} \frac{2n+1}{4\pi} \hat{\psi}(n) P_n(t), \quad t \in [-1, 1]$$

P_n ... n -th Legendre polynomial

The mollifier method

(Louis & Maas, 1990)

We compute for some mollifier $\psi : [-1, 1] \rightarrow \mathbb{R}$

$$\begin{aligned} \psi \star f(\xi) &= \int_{\mathbb{S}^2} \psi(\langle \xi, \eta \rangle) f(\eta) \, d\eta \\ &= \sum_{n=0}^{\infty} \sum_{k=-n}^n \hat{\psi}(n) \hat{f}(n, k) Y_n^k(\xi), \quad \xi \in \mathbb{S}^2 \end{aligned}$$

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Assumptions for discrete noisy data

- ▶ Given data

$$g(\xi_m) = \mathcal{R}f(\xi_m) + \varepsilon(\xi_m)$$

for $m = 1, \dots, M$.

- ▶ We have to make some assumptions for the noise ε , i.e.

White Gaussian noise

- ▶ unbiased: $\mathbb{E}\varepsilon(\xi_m) = 0$
- ▶ uncorrelated: $\mathbb{E}[\varepsilon(\xi_m)\varepsilon(\xi_l)] = 0$
- ▶ variance: $\mathbb{E}[\varepsilon(\xi_m)^2] = \sigma^2$

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How to measure the error?

- Mean integrated squared error (**MISE**)

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- For $s, S \geq 0$, define the function class

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Variance error

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$$\mathbb{E} \|\mathcal{E}_{N,\psi}\|_{L^2}^2 = \frac{4\pi\sigma}{M} \|\mathcal{R}^\dagger \psi\|_{L^2}^2 = \frac{4\pi\sigma}{M} \sum_{n=0}^N \sum_{k=-n}^n \left| \frac{\hat{\psi}(n)}{\hat{\mathcal{R}}(n)} \right|^2$$

σ ... *variance of the data error* ε

M ... *number of data points*

- To minimize the variance, the coefficients $\hat{\psi}(n)$ should decrease for $n \rightarrow \infty$

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Bias error

$$\|f - \psi \star \mathcal{R}^\dagger \mathcal{L}_N(\mathcal{R}f)\|_{L^2} \leq \underbrace{\|f - \psi \star f\|}_{\text{smoothing error}} + \underbrace{\|\psi \star (f - \mathcal{R}^\dagger \mathcal{L}_N \mathcal{R}f)\|}_{\text{discretization error}}$$

► Smoothing error

$$\begin{aligned} \|f - \psi \star f\|^2 &= \sum_{n=0}^{\infty} \sum_{k=-n}^n \left| 1 - \hat{\psi}(n) \right|^2 \left| \hat{f}(n, k) \right|^2 \\ &\leq \max_{n \in \mathbb{N}} \frac{\left| 1 - \hat{\psi}(n) \right|^2}{\left(n + \frac{1}{2} \right)^{2s}} \|f\|_s^2 \end{aligned}$$

- To minimize this, the coefficients $\hat{\psi}(n)$ should be close to zero
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Bias error

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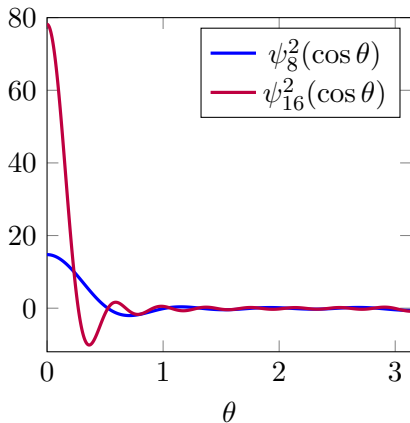
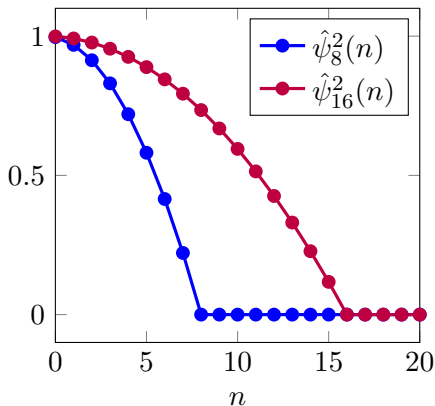
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Asymptotic optimality of ψ_L^s

Theorem

- ▶ Let $s > \frac{3}{2}$ and $S > 0$
- ▶ For every $N \in \mathbb{N}$ let the hyperinterpolation $\mathcal{L}_N$ be exact and have constant weights and the number of points $M \sim N^2$
- ▶ Let the data error $\varepsilon(l, m)$ be i.i.d. with expected value 0 and variance δ^2

Then the family $\{\psi_L^s\}_{L \in \mathbb{R}_+}$ is asymptotically optimal for the class $\mathcal{F}(s, S)$.
 For $N \rightarrow \infty$ there exist parameters $L(N)$ such that

$$\sup_{f \in \mathcal{F}(s, S)} \mathbb{E} \|f - \mathcal{E}_{N, \psi_{L(N)}^s}(\mathcal{R}f + \varepsilon)\|_{L^2}^2$$

$$\simeq \inf_{\psi} \sup_{f \in \mathcal{F}(s, S)} \mathbb{E} \|f - \mathcal{E}_{N, \psi}(\mathcal{R}f + \varepsilon)\|_{L^2}^2 \simeq \text{const} \cdot N^{\frac{-s}{s+3/2}}.$$

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1. Introduction

Definition

History

2. The Funk–Radon transform

Spherical harmonics as eigenfunctions of the Funk–Radon transform

Sobolev spaces

3. Inversion of the Funk–Radon transform

Discrete data

Noisy data

Bias–variance tradeoff

Optimal mollifiers

4. Generalizations of the Funk–Radon transform

Convolution operators

Means along circles on the sphere

Convolution operators

- Recall the Funk–Radon transform

$$\mathcal{R}f(\xi) := \frac{1}{2\pi} \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, ds(\eta), \quad \xi \in \mathbb{S}^2$$

- The **spherical convolution** with $g : [-1, 1] \rightarrow \mathbb{R}$ is defined as

$$g \star f(\xi) := \int_{\mathbb{S}^2} f(\eta) g(\langle \xi, \eta \rangle) \, d\eta, \quad \xi \in \mathbb{S}^2$$

- The Funk–Radon transform is the convolution with the delta distribution
- Such convolutions can also be expressed via spherical harmonics

$$\widehat{g \star f}(n, k) = \hat{g}(n) \hat{f}(n, k)$$

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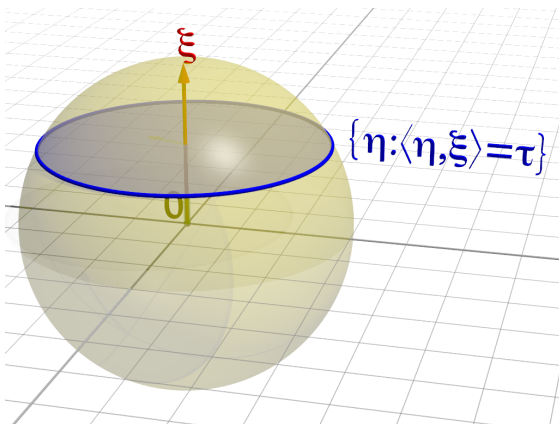
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- For fixed $\tau \in (-1, 1)$, the **translation operator**

(Rudin, 1950)

$$\mathcal{T}_\tau f(\xi) := \frac{1}{2\pi\sqrt{1-\tau^2}} \int_{\langle \eta, \xi \rangle = \tau} f(\eta) \, ds(\eta), \quad \xi \in \mathbb{S}^2$$

- “All circles with a fixed radius”

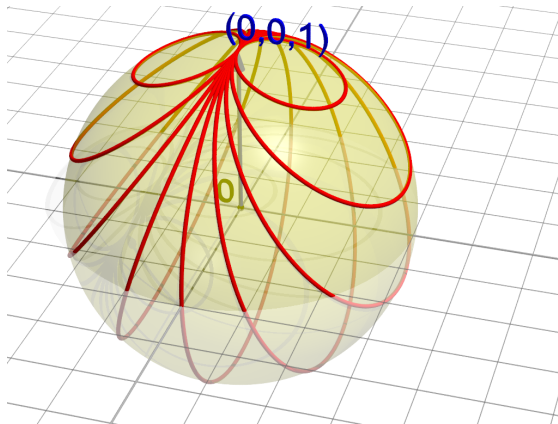


► The **spherical slice transform**

(Helgason, 1999)

$$\mathcal{T}_t f(\xi) := \frac{1}{2\pi\sqrt{1-t^2}} \int_{\langle \eta, \xi \rangle = \langle \xi, e_3 \rangle} f(\eta) \, ds(\eta), \quad \xi \in \mathbb{S}^2$$

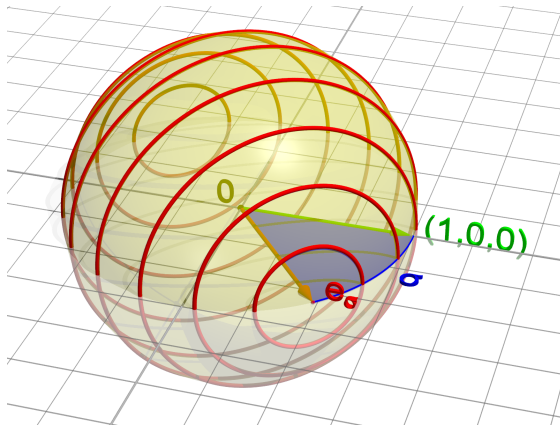
- “All circles passing through the north pole e_3 ”



All circles perpendicular to the z -axis

(Gindkin et. al., 1993)

$$\mathcal{T}f(\sigma, t) := \frac{1}{2\pi\sqrt{1-t^2}} \int_{\langle \eta, e_\sigma \rangle = t} f(\eta) \, ds(\eta), \quad \xi \in \mathbb{S}^2$$





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